Foundation of AI and Automation

Assignment 2:

Instructions: Answer all the following questions in your own words. Each question carries equal marks.

Q1. What is an agent in Artificial Intelligence? Give one real-life example of an agent.

Q2. Explain the term "environment" in AI. List any four properties (nature) of an environment.

Q3. Differentiate between a simple reflex agent and a goal-based agent (any two points).

Q4. What is a problem-solving agent? Write the basic steps involved in problem formulation.

Q5. What is a search algorithm? Why is searching important in Artificial Intelligence?

Q6. Explain Breadth First Search (BFS) with a simple example. Write one advantage and one disadvantage of BFS.

Q7. Explain Depth First Search (DFS) with a simple example. How is it different from Brute Force Search?

Q8. What is a heuristic function? Explain the working of the Hill Climbing algorithm in brief.

Q9. Explain the Best First Search algorithm and the A* algorithm. What is the main difference between A* and AO* algorithm?

Q10. What is Knowledge Representation in AI? List any three techniques used for representing knowledge.

Note for Students:

- Answers should be written in simple, clear language with diagrams/examples wherever possible.
- Suggested length: 3–5 lines per answer (except numericals/algorithm-based questions).